AP Computer Science A
Ticket Out the Door
Set 5: Methods of the Math Class

Name _____ Period _____

Skill 5.1 Exercise 1

Complete each task independently. Use Java statements and write exact output where requested.

(a) Assume x is a nonnegative double that has already been initialized. Declare a double variable y and store the square root of x in it.

(b) Use Math.pow to declare and initialize a double variable result with the value of 3 raised to the fourth power.

(c) Explain why this statement does not compile. Correct it so the complete absolute value, including the fractional part, is stored in k.
int k = Math.abs(-127.5);

(d) Write the exact output of each statement. Include .0 when Java prints it.

(i) System.out.println(Math.min(100, 200));
Output: _____

(ii) System.out.println(Math.max(100.0, 200));
Output: _____

(iii) System.out.println(Math.max(Math.max(100, 200), 50));
Output: _____

(iv) System.out.println(Math.sqrt(81));
Output: _____

(v) System.out.println(Math.round(-157.2));
Output: _____

(vi) System.out.println(Math.ceil(-157.7));
Output: _____

(vii) System.out.println(Math.floor(-157.7));
Output: _____

(viii) System.out.println(Math.ceil(157));
Output: _____

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Skill 5.2 Exercise 1	
Use Math.random() to declare and initialize one variable per row with the stated type and range.	
double r: 0.0 <= r < 1.0 0 included; 1 excluded.	
int card: 1 through 52 Both endpoints included.	
int number: 100 through 199 100 included; 200 excluded.	
int offset: -100 through 0 Both endpoints included.	
Explain why (int)Math.random() * 6 + 1 always produces 1. Write a corrected expression that can produce any integer from 1 through 6.	

Skill 5.3 Exercise 1
Translate each expression into Java. Declare a double variable with the indicated name; you do not need to calculate its decimal value.
(a) d1 = 3π sin(187°) + |cos(122°)| Use Math.PI, Math.sin, Math.cos, and Math.abs. Convert each angle with Math.toRadians because sin and cos expect radians.
(b) d2 = (14.72 raised to the power 3.801) + ln(72) Use Math.pow and Math.log. The notation ln means natural logarithm (base e).

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Skill 5.3 Exercise 2
A game rolls two six-sided dice. Each die can show any integer from 1 through 6, inclusive. The program calculates the distance between the two rolls and a score. Use the variables from earlier parts when completing later parts
(a) Declare and initialize int variables die1 and die2. Use a separate call to Math.random() for each die. Every face from 1 through 6 must be possible, with approximately equal likelihood.
(b) Declare and initialize an int variable difference using Math.abs so it stores the nonnegative difference between the rolls. For example, rolls of 2 and 5 produce a difference of 3.
(c) The score is the square root of the sum of the squares of the two rolls. Use Math.pow and Math.sqrt to declare and initialize a double variable score. Then print difference and score on separate lines.
(d) Suppose die1 is 3 and die2 is 4. Write the two lines printed by part (c). Explain why score must be stored as a double even when its value is a whole number.

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